THREE DESCRIPTIONS OF THE DEPTH OF A FINITE MODULE OVER AN IDEAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The placement of the ideal in an associated prime when the depth against it is zero. Invoked in the base case (source lines 1941-1945). Produces a nonzero homomorphism out of the quotient by the ideal.
- The description of local cohomology as the colimit of the Ext modules out of the quotients by the powers of the ideal. Invoked at the end of the counted proof (source lines 1989-1990). Compares the vanishing of local cohomology with the vanishing of those Ext modules.
- The finiteness of Ext modules out of a finite module over the quotient. Invoked inside the induction step (source lines 1957-1959). Turns the long exact sequence into short exact sequences by Nakayama.
- The invariance of the depth under replacing the ideal by its powers. Invoked inside the counted proof (source line 1971). Extends the vanishing statement to all powers of the ideal.

Source reference. Original reference algebra-lemma-annihilate-ext: sources/stacks-dualizing/pinned-original-source.tex, source lines 1959–1959. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-annihilate-ext, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference algebra-lemma-ass-zero-divisors: sources/stacks-dualizing/pinned-original-source.tex, source lines 1943–1943. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-ass-zero-divisors, which belongs to another chapter of the project and is not part of this card. The

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Source reference. Original reference algebra-lemma-depth-finite-noetherian: sources/stacks-dualizing/pinned-original-source.tex, source lines 1938–1938. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-depth-finite-noetherian, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference algebra-lemma-ext-noetherian: sources/stacks-dualizing/pinned-original-source.tex, source lines 1958–1958. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-ext-noetherian, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference algebra-lemma-regular-sequence-powers: sources/stacks-dualizing/pinned-original-source.tex, source lines 1972–1972. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-regular-sequence-powers, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference algebra-lemma-silly: sources/stacks-dualizing/pinned-original-source.tex, source lines 1944–1944. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-silly, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-local-cohomology-ext: sources/stacks-dualizing/pinned-original-source.tex, source lines 1308–1308. Navigation pointer only: the counted text refers to the Stacks item lemma-local-cohomology-ext, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let A be a Noetherian ring, let $I \subset A$ be an ideal, and let M be a finite A-module such that $IM \neq M$. Then the following integers are equal:*

1. *$\text{depth}_I(M)$,*
2. *the smallest integer i such that $\text{Ext}^i_A(A/I, M)$ is nonzero, and*
3. *the smallest integer i such that $H^i_I(M)$ is nonzero.*

Moreover, we have $\text{Ext}^i_A(N, M) = 0$ for $i < \text{depth}_I(M)$ for any finite A-module N annihilated by a power of I.

Proof. We prove the equality of (1) and (2) by induction on $\text{depth}_I(M)$ which is allowed by Algebra, Lemma algebra-lemma-depth-finite-noetherian.

Base case. If $\text{depth}_I(M) = 0$, then I is contained in the union of the associated primes of M (Algebra, Lemma algebra-lemma-ass-zero-divisors). By prime avoidance (Algebra, Lemma algebra-lemma-silly) we see that $I \subset \mathfrak{p}$ for some associated prime $\mathfrak{p}$. Hence $\text{Hom}_A(A/I, M)$ is nonzero. Thus equality holds in this case.

Assume that $\text{depth}_I(M) > 0$. Let $f \in I$ be a nonzerodivisor on M such that $\text{depth}_I(M/fM) = \text{depth}_I(M) - 1$. Consider the short exact sequence

$$0 \to M \to M \to M/fM \to 0$$

and the associated long exact sequence for $\text{Ext}^*_A(A/I, -)$. Note that $\text{Ext}^i_A(A/I, M)$ is a finite A/I-module (Algebra, Lemmas algebra-lemma-ext-noetherian and algebra-lemma-annihilate-ext). Hence we obtain

$$\text{Hom}_A(A/I, M/fM) = \text{Ext}^1_A(A/I, M)$$

and short exact sequences

$$0 \to \text{Ext}^i_A(A/I, M) \to \text{Ext}^i_A(A/I, M/fM) \to \text{Ext}^{i+1}_A(A/I, M) \to 0$$

Thus the equality of (1) and (2) by induction.

Observe that $\text{depth}_I(M) = \text{depth}_{I^n}(M)$ for all $n \geq 1$ for example by Algebra, Lemma algebra-lemma-regular-sequence-powers. Hence by the equality of (1) and (2) we see that $\text{Ext}^i_A(A/I^n, M) = 0$ for all n and $i < \text{depth}_I(M)$. Let N be a finite A-module annihilated by a power of I. Then we can choose a short exact sequence

$$0 \to N' \to (A/I^n)^{\oplus m} \to N \to 0$$

for some $n, m \geq 0$. Then $\text{Hom}_A(N, M) \subset \text{Hom}_A((A/I^n)^{\oplus m}, M)$ and $\text{Ext}^i_A(N, M) \subset \text{Ext}^{i-1}_A(N', M)$ for $i < \text{depth}_I(M)$. Thus a simply induction argument shows that the final statement of the lemma holds.

Finally, we prove that (3) is equal to (1) and (2). We have $H^p_I(M) = \text{colim}\, \text{Ext}^p_A(A/I^n, M)$ by Lemma lemma-local-cohomology-ext. Thus we see that $H^i_I(M) = 0$ for $i < \text{depth}_I(M)$. For $i = \text{depth}_I(M)$, using the vanishing of $\text{Ext}^{i-1}_A(I/I^n, M)$ we see that the map $\text{Ext}^i_A(A/I, M) \to H^i_I(M)$ is injective which proves nonvanishing in the correct degree. $\square$